

Refrigeration Simulation Report

Operating Conditions

Parameter	Value
compressor_efficiency	85
condenser_temp	40
evaporator_temp	-15
subcool	5
superheat	5

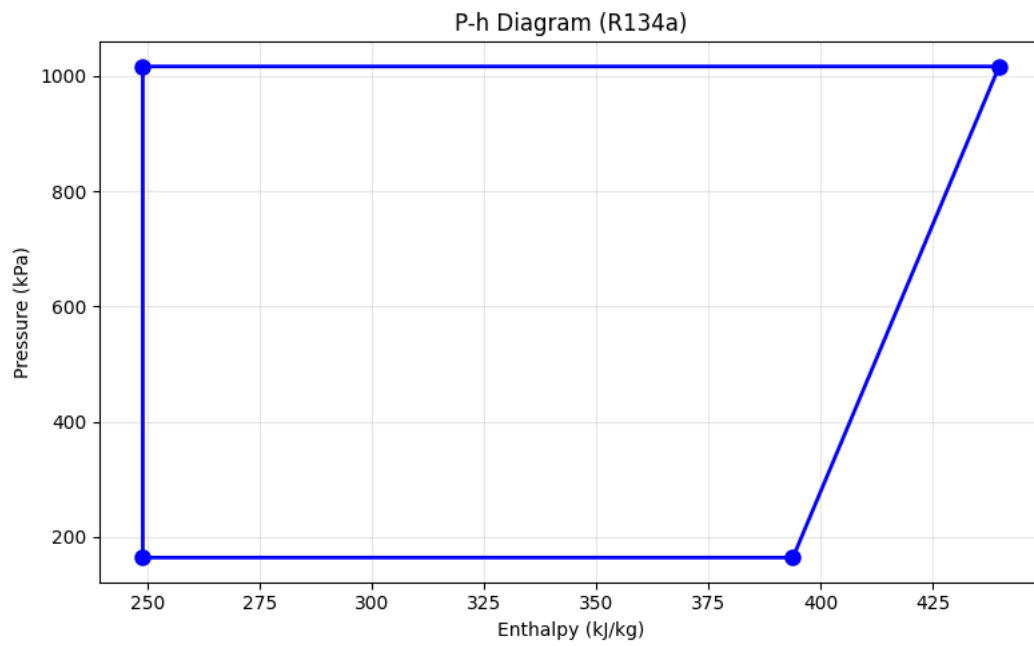
Performance

Metric	Value
COP	3.152
compressor_power	3.172
compressor_work	45.94
cooling_capacity	10
heat_rejected	190.74
mass_flow_rate	0.06906
refrigeration_effect	144.8
volumetric_efficiency	85

State Points

Point	P (kPa)	T (°C)	h (kJ/kg)	s	quality
1	163.94	-10.0	393.80	1.753	N/A
2	1016.59	58.6	439.73	1.774	N/A
3	1016.59	35.0	248.99	1.167	N/A
4	163.94	-15.0	248.99	1.192	0.329

P-h Diagram



T-s Diagram

